

## KARST AND BIODIVERSITY

One of the most important contributions of Europe to the Earth's biodiversity is well hidden below its surface – an exceptionally rich subterranean fauna, with its top diversity hotspots in Dinaric Karst (from East Italy, Slovenia, Croatia, Bosnia and Herzegovina to Montenegro) and Pyrenees.



Planinska jama, Slovenia



Crnulja Cave, Herzegovina

## Vulnerability of karst groundwater

Conservation of *Proteus* and other endemic cave animals is not only an attempt to preserve the subterranean biodiversity, it is among the top concerns of mankind: with cave animals we share the most important resource of fresh water – groundwater.

About 75% of EU residents depend on groundwater for their water supply. Large part of groundwater is stored in karst, which underlie nearly 35% of Europe, where many important cities, including Bristol, London, Paris, Rome and Vienna are largely dependent on karst groundwater. In Slovenia, groundwater is serving nearly all water needs.



Rubbish dumped in caves



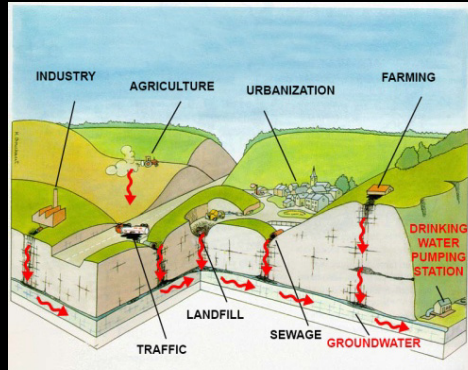
Overuse of biogas slurry

Karst areas are the world's most vulnerable landscapes to environmental impacts. All toxic waste dumped on the surface is washed by precipitation into groundwater, very difficult to remediate once contaminated.

## WHY IS *PROTEUS* VULNERABLE?

*Proteus* is threaten mostly by raising pollution of karst groundwater. Due to its longevity (up to 100 years) even low concentrations of pollutants may accumulate to lethal values in *Proteus* body and habitat.

Recent negative anthropogenic influences come from intensive agriculture, energy production (karst water damming), and unregulated urbanization (groundwater pumping and sewage disposal).



In Slovenia, *Proteus* has been protected since 1922. Listed in the EU Habitats Directive, Bern and Ramsar Convention, the IUCN Red List of Threatened Species, *Proteus* is one of EU priority species in need of strict protection, including the Natura 2000 network. However, *Proteus* still has no monitoring scheme or conservation action plan.

## SOS *Proteus*

The Tular Cave Laboratory also serves as a sanctuary for *Proteus* accidentally washed-out of their subterranean habitat during seasonal flooding. Some *Proteus* are found stranded on fields and may successfully be returned to their source populations. Other are injured, and a veterinary inspection and animal care is conducted by Dr. Zlatko Golob (Sanctuary for protected animal species Golob d.o.o.).



Petelinje Jezero, 2014



Lož, 2015

# With *Proteus* we share dependence on groundwater



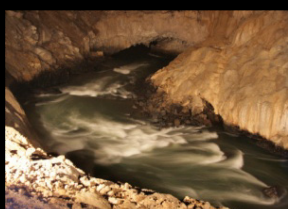
## 55 years of Tular Cave Laboratory

## *Proteus anguinus* [olm, human fish, človeška ribica]

One of the world's prime symbols of natural heritage and its history of research, is the olm (*Proteus anguinus*).



This blind amphibian, a dweller of the subterranean rivers, is strictly endemic to the Dinaric Karst. It is the only European cave vertebrate, and by far the largest cave animal in the world. *Proteus* is a relic species of a rich Tertiary fauna, and has no living close relatives.



Subterranean rivers of Dinaric Karst

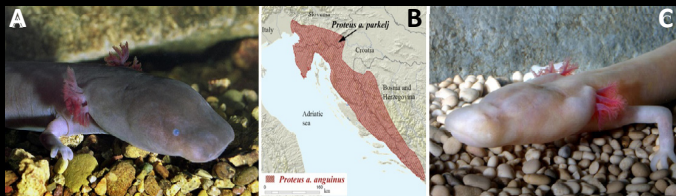


Inaccessible habitat of *Proteus*

Around 300 localities (partly interconnected caves or cave springs) of *Proteus* have been recorded, of which 200 are in Slovenia. In some localities *Proteus* already become extinct.

## *Proteus anguinus parkelj* [black *Proteus*]

The morphologically most distinct of all *Proteus* populations, discovered in 1986, is limited to a single cave system in Bela Krajina (South East Slovenia, on less than 30 km<sup>2</sup>). Even a single local pollution incident could have a devastating impact on the entire population.



A) Black *Proteus*, C) white *Proteus*, and B) their distribution (map after IUCN List of Red List of Threatened Species)

Since 2002, a colony of the extremely endangered pigmented subspecies *Proteus anguinus parkelj* (the black *Proteus*) is carefully studied in Tular Cave Laboratory.

## TULAR CAVE LABORATORY

Since only small fragments of the subterranean habitat can be accessed by man, most knowledge about *Proteus* has been acquired in laboratories, away from its natural environment. One such case is the Tular Cave Laboratory.

In 1960, the cave was turned into a subterranean laboratory by speleobiologist Marko Aljančič (1933–2007), who populated it with *Proteus*.



Marko Aljančič observing *Proteus*, 1962



Successful ex-situ breeding of *Proteus*

## 55 years of conservation and research on *Proteus*

Tular is the only cave laboratory in Slovenia, and – apart from the Subterranean Laboratory of CNRS in Moulis, France – the only place with successful ex-situ breeding program of this highly endangered cave amphibian. Since 2007, the Laboratory and the Society have been led by Gregor Aljančič.



Laboratory chamber with pools



Semi-natural habitat of *Proteus*

In the laboratory, the ecology and behaviour of *Proteus* are studied, focused mainly around its breeding. The research is based on observations, carefully designed not to harm or stress the animals.

## INFO

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## Conservation of *Proteus* and public education



The conservation of *Proteus* and its subterranean habitat may truly be effective only by education and constant public outreach, emphasizing vulnerability of karst groundwater: the habitat of *Proteus* is the ultimate source of drinking water.

## On the trace of environmental DNA (eDNA)



eDNA is constantly washed away



Collecting water samples of *Proteus* eDNA

The Laboratory is introducing a forensic method to detect molecular traces of *Proteus* in groundwater. The non-invasive, DNA-based method was successfully applied to monitor the distribution of *Proteus*. Utilizing the eDNA, new localities of the extremely rare black *Proteus* were discovered.

Environmental DNA method is one step towards the urgently needed monitoring scheme for *Proteus* populations and their habitat.

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